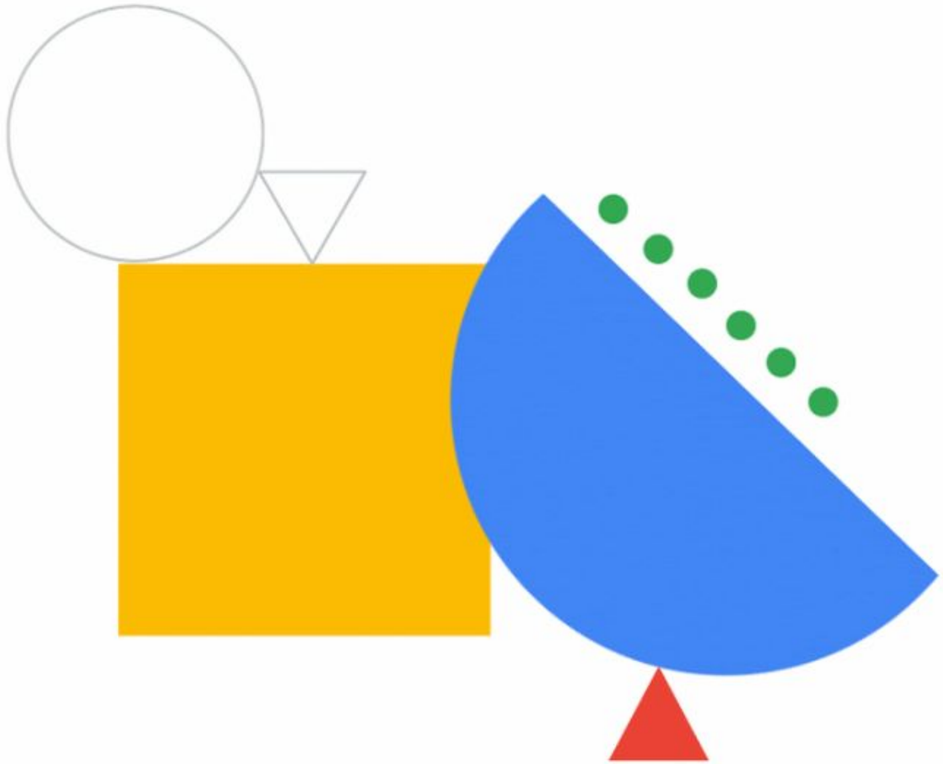
Google Cloud

Monitoring Google Cloud Network



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Objectives

01 Collect and analyze VPC Flow Logs, Firewall Rules Logging, load balancer logs, and Cloud NAT logs.

02 Enable and monitor Packet Mirroring.

03 Explain the capabilities of the Network Intelligence Center.



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In this section, you explore



- VPC Flow Logs
- Firewall Rules Logging
- Load balancer logs
- Cloud NAT logs
- Packet Mirroring
- Network Intelligence Center

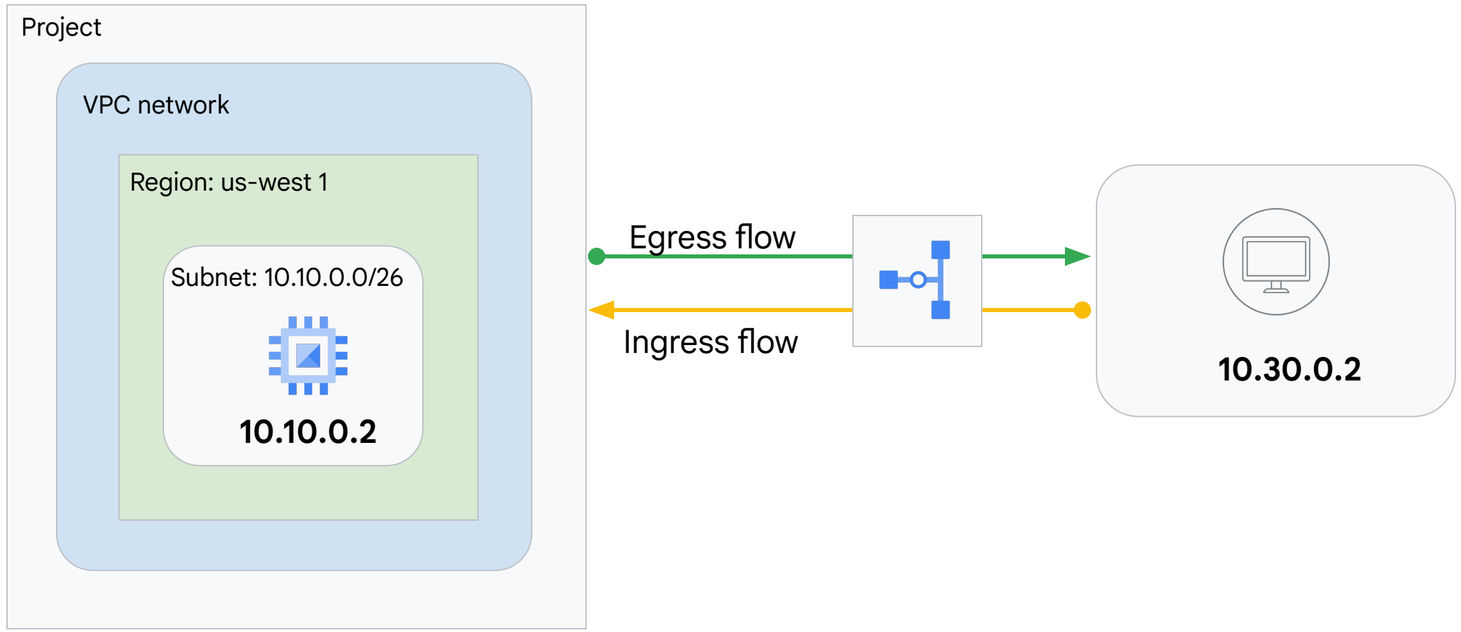
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VPC Flow Logs is used to monitor network by recording a portion of network flows sent and received by VM instances (including GKE nodes).

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VPC Flow Logs example

VM to external traffic flow



Flow logs reported from the VM:

Egress flow: 10.10.0.2 to 10.30.0.2

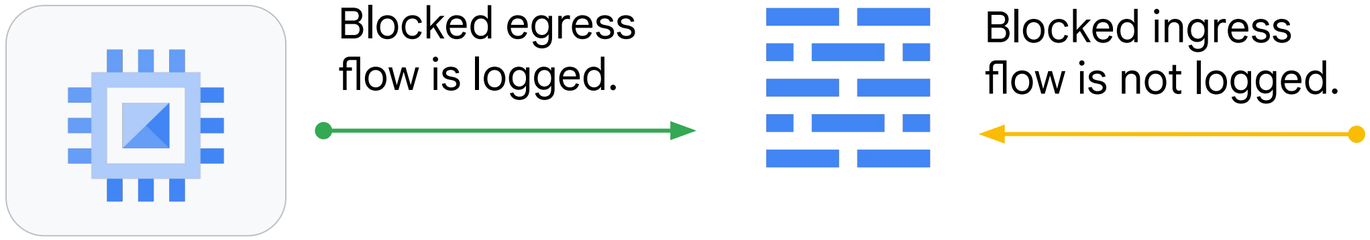
Ingress flow: 10.30.0.2 to 10.10.0.2

The VM and and on-premises endpoint are connected through a Cloud VPN or Cloud Interconnect instance

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VPC Flow Logs properties

- Samples are from the VM's perspective.
  - An egress deny firewall is logged.
  - An ingress deny firewall rule is not logged.
- Samples are logged for each VMs. This includes inbound and outbound traffic:
  - VM to VM, VM to a host on the internet, VM to on-premises host, etc.



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Enable VPC Flow Logs per VPC subnet

VPC Flow Logs is activated or deactivated at a subnet level.

All VMs within that subnet have VPC Flow Logs automatically enabled.

You can enable VPC Flow Logs during subnet creation.

You can optionally adjust log sampling and aggregation to adjust the metadata and sample rate written to logs.

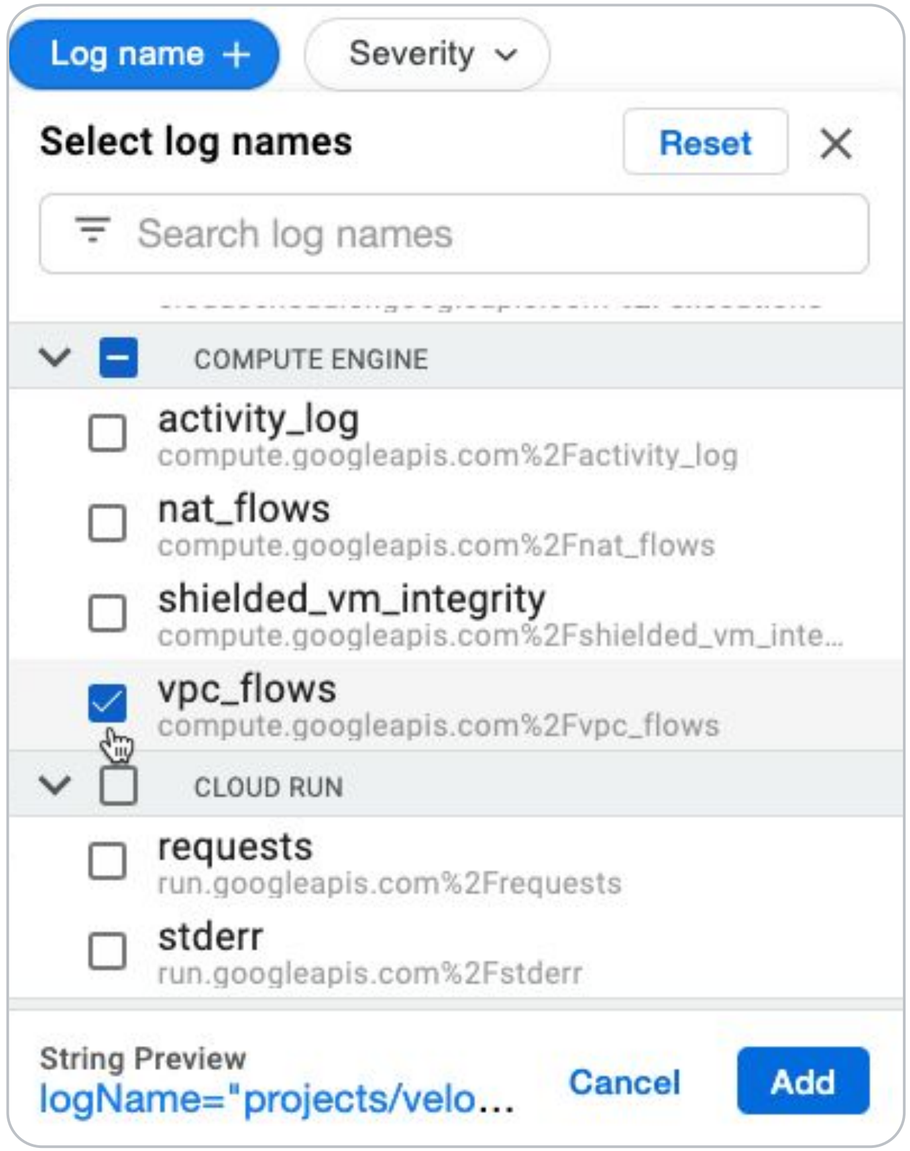
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Log entries contain many useful fields

Field	Type	Description
src_ip	string	Source IP address
src_port	int32	Source port
dest_ip	string	Destination IP address
dest_port	int32	Destination port
protocol	int32	IANA protocol number

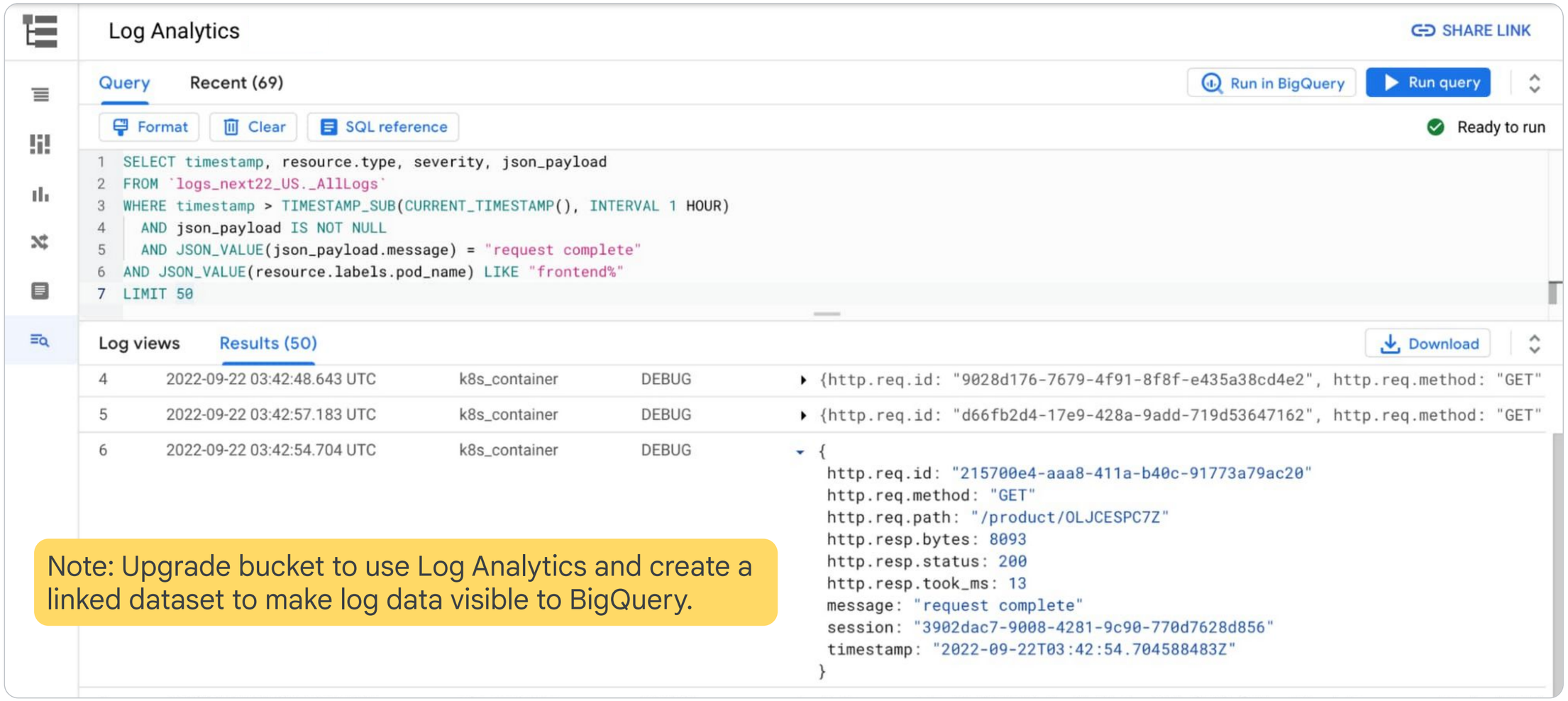
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Use Logging to analyze your VPC Flow Logs



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Analyze logs with Log Analytics



Note: Upgrade bucket to use Log Analytics and create a linked dataset to make log data visible to BigQuery.

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Firewall Rules Logging



Did my firewall rules cause that application outage?



How many connections match the rule I just created?

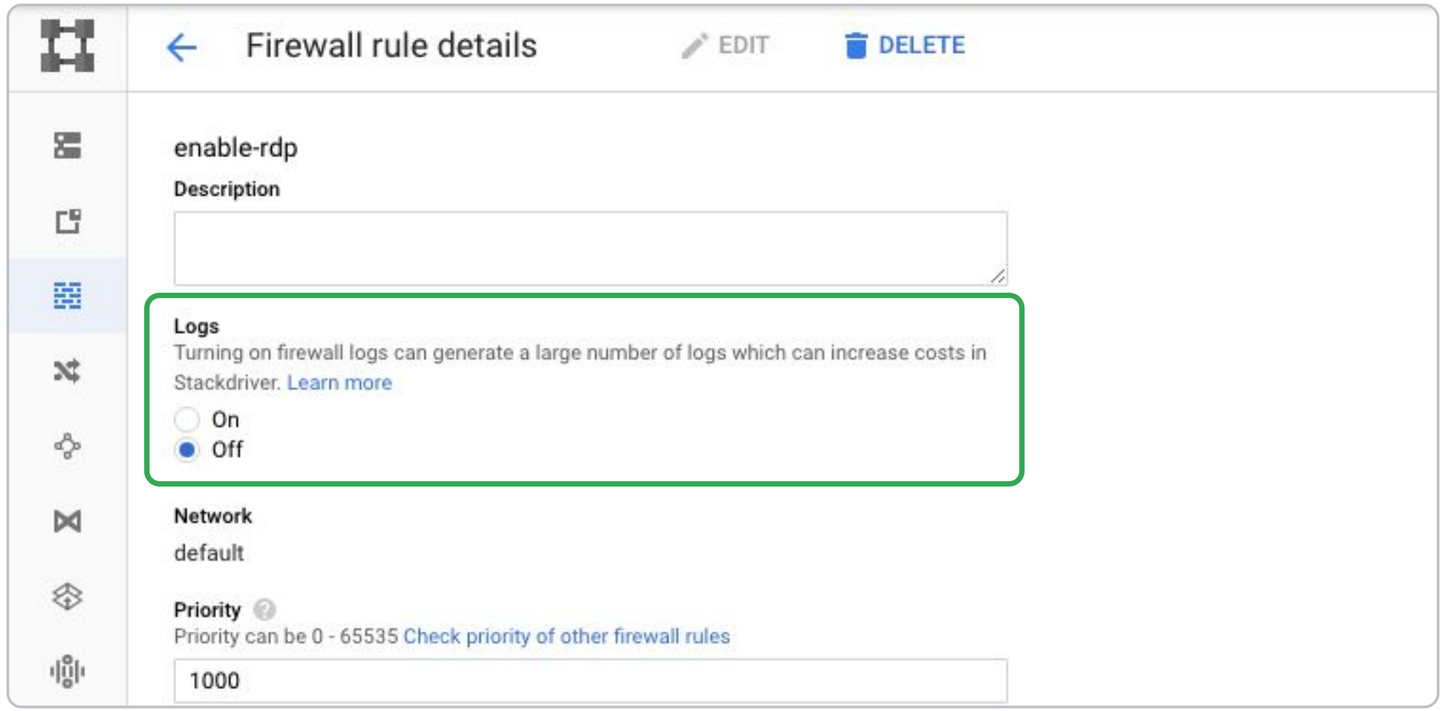


Are my firewall rules stopping (or allowing) the correct traffic?

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Enabling Firewall Rules Logging in the console

- Firewall Rules Logging is **disabled** by default.
- You enable it on a per-rule basis.



Note: Firewall Rules Logging can only record TCP and UDP connections. For other protocols, use Packet Mirroring.

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Enabling Firewall Rules Logging in the CLI

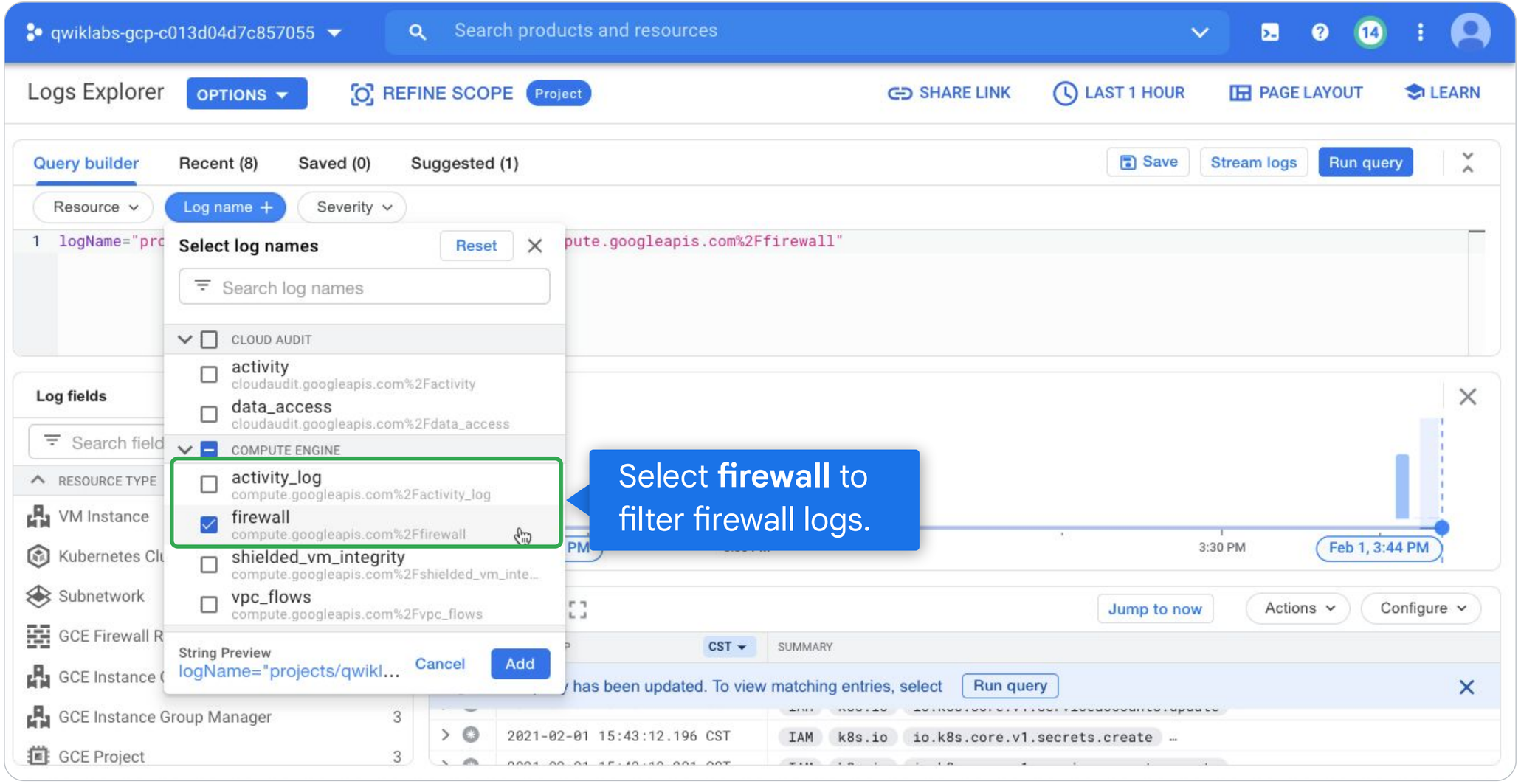
Firewall Rules Logging can also be enabled or disabled by using the following **gcloud** CLI commands. Substitute [NAME] for the name of your firewall rule.

```
Activate:
gcloud compute firewall-rules update [NAME] --enable-logging

Deactivate:
gcloud compute firewall-rules update [NAME] --no-enable-logging
```

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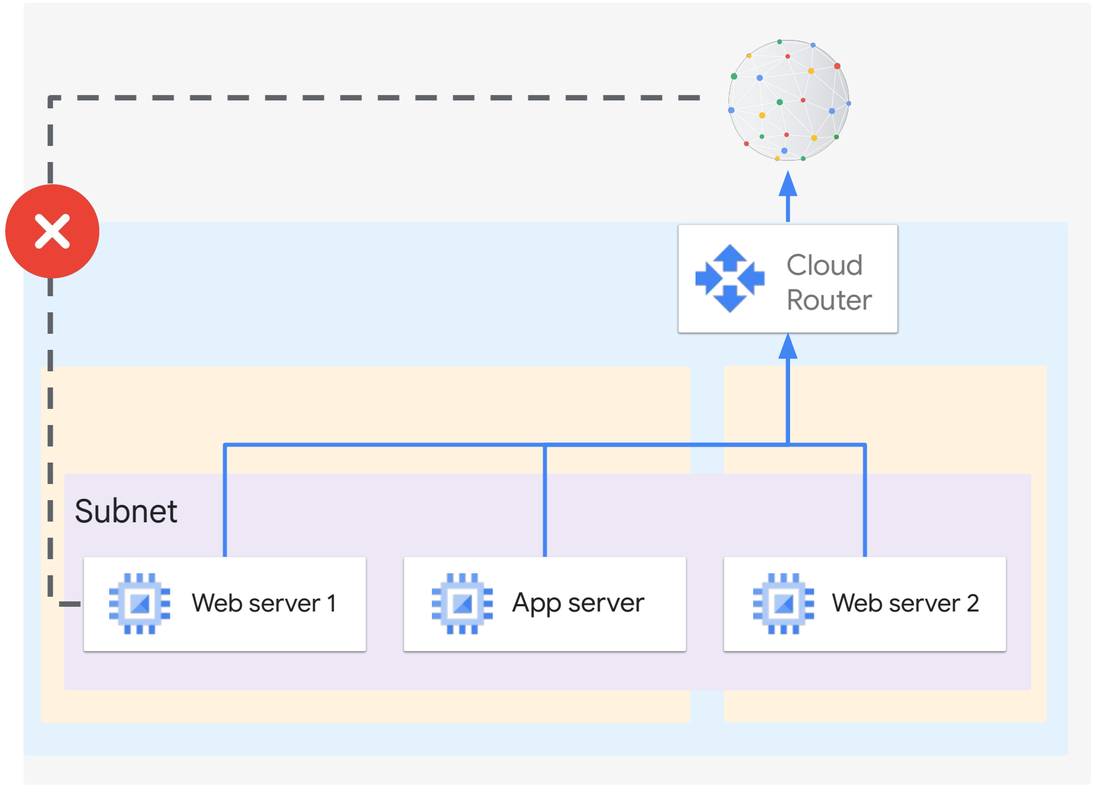
Viewing the Firewall Rules logs



Google Cloud

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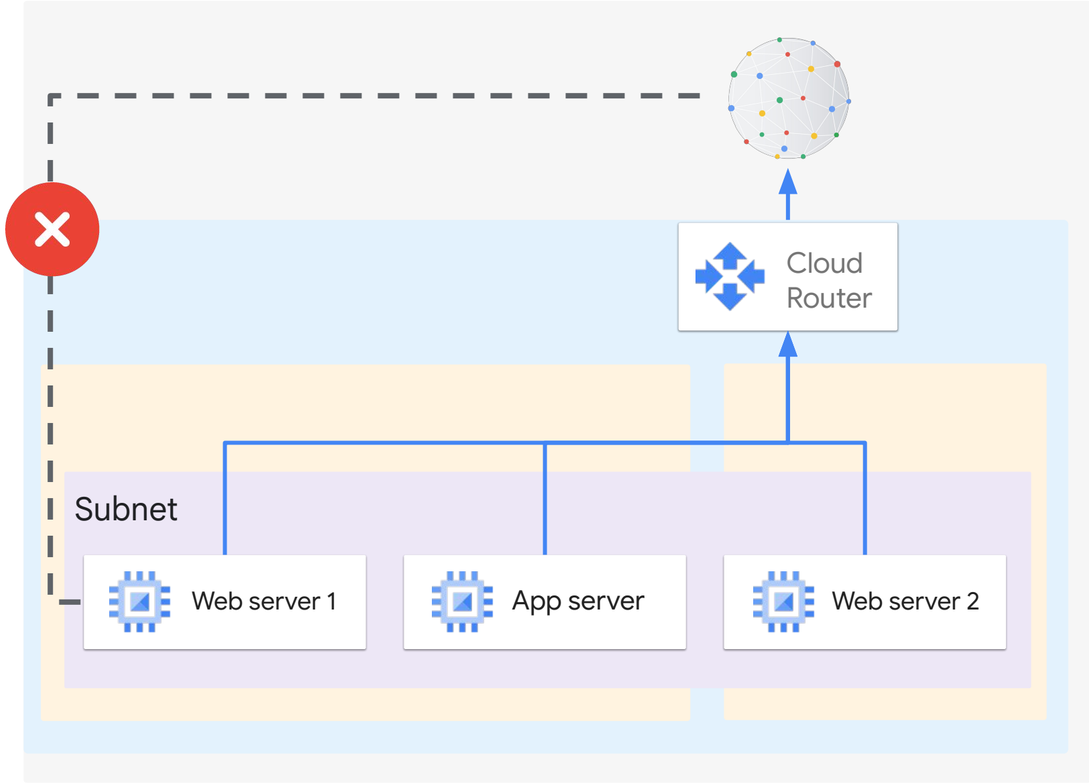
Firewall rules provide microsegmentation



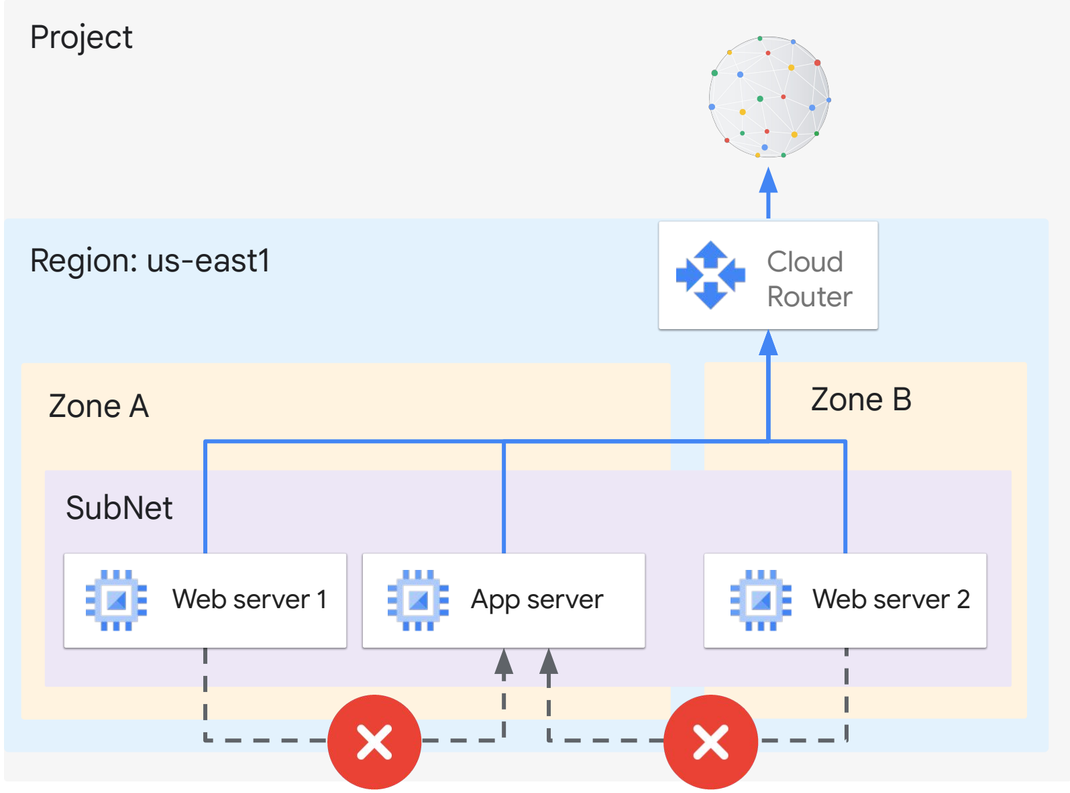
Segmentation or gateway-centric

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Firewall rules provide microsegmentation



Segmentation or gateway-centric

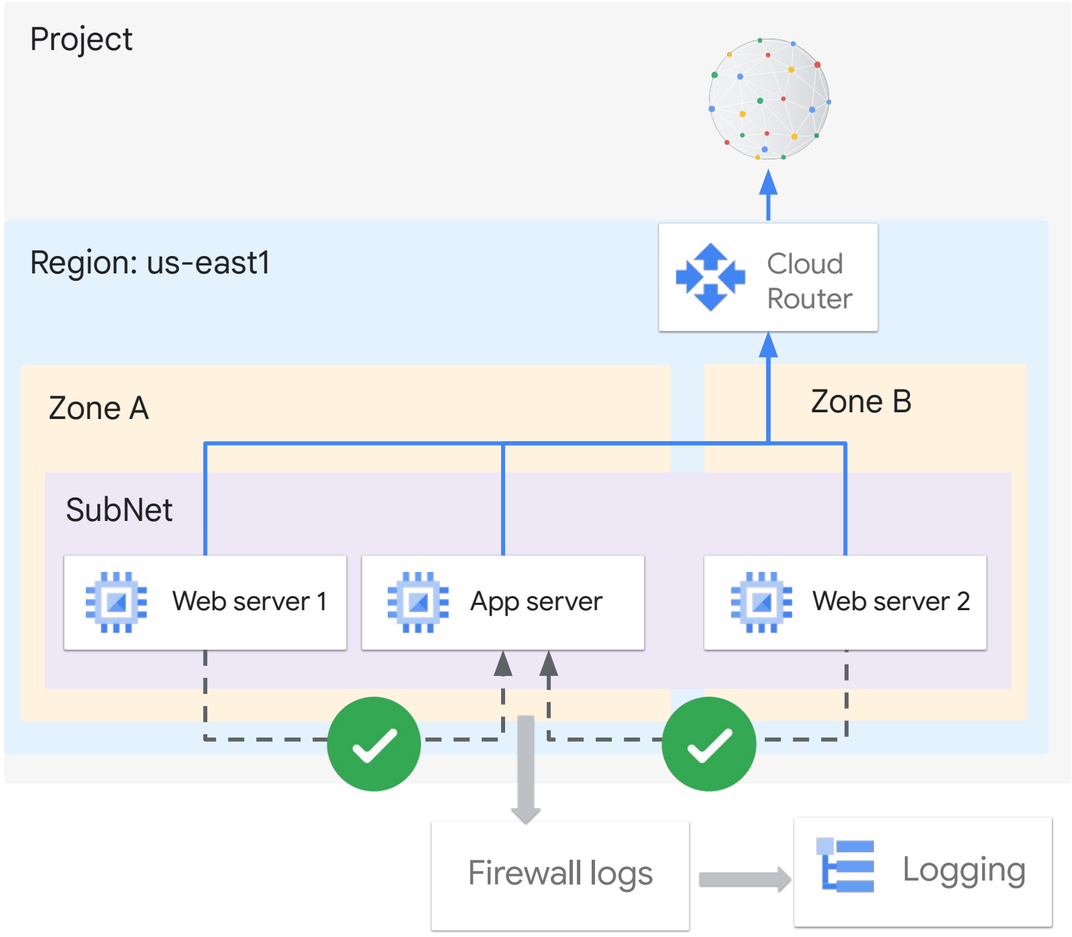


Microsegmentation or VM-centric

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Troubleshooting: Using rules to catch incorrect traffic

- Logging all denied connections will create too many log entries.
- Temporarily create a high-priority rule (low-priority value) to allow traffic to the server. Enable Cloud Logging.
- If traffic now gets through, examine the logs to find the root cause.



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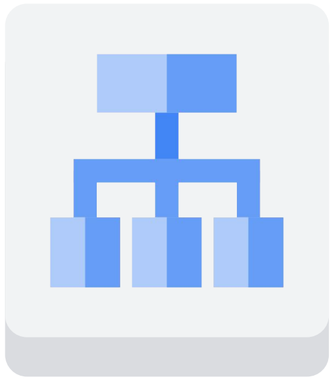
In this section, you explore



- VPC Flow Logs
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- Load balancer logs
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Load balancer support for Cloud Logging



- All the Google Cloud load balancers support Cloud Logging and Cloud Monitoring:
  - Internal and external Application Load Balancers
  - Internal and external Network Load Balancers
  - Internal and external Proxy Load Balancers
- The log type, log fields, and metrics supported vary based on the the load balancer type.
- Load balancing logs can be used to debug and analyze user traffic.

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The internal and external Application Load Balancers support logging

- **Activated and deactivated on a per backend service basis**
  - For external Application Load Balancers with backend buckets, logging is automatically enabled and cannot be deactivated.
  - Logging can be enabled on a per backend service basis.
  - URL map might reference more than one backend service.
  - Use exclusion, if you do not want the logs to be stored in Cloud Logging.

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Fields in a log record

- `LogEntry`: Contains severity, project ID, project number, and timestamp information.
- `httpRequest`: Contains a method, URL, status, remote IP address, latency string.
- `resource`: Contains the monitored resource associated with a log entry.
- `jsonPayload`: Contains `statusDetails field` that includes a string that explains why the load balancer returned the HTTP status, cache, and failure information.
  - **Redirects** (HTTP response status code 302 found) issued from the load balancer are not logged. Redirects issued from the backend instances are logged.

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Example of using a load balancing log record

- **Error:** Load balancer generates an `HTTP error response code (5XX)`.
- Refer to Load balancer logs to determine the source of error.
  - **`statusDetails field: response_sent_by_backend`** field indicates it is a backend issue.
  - **`statusDetails field: failed_to_pick_backend`** field indictes that the load balancer failed to pick a healthy backend to handle a request.

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In this section, you explore

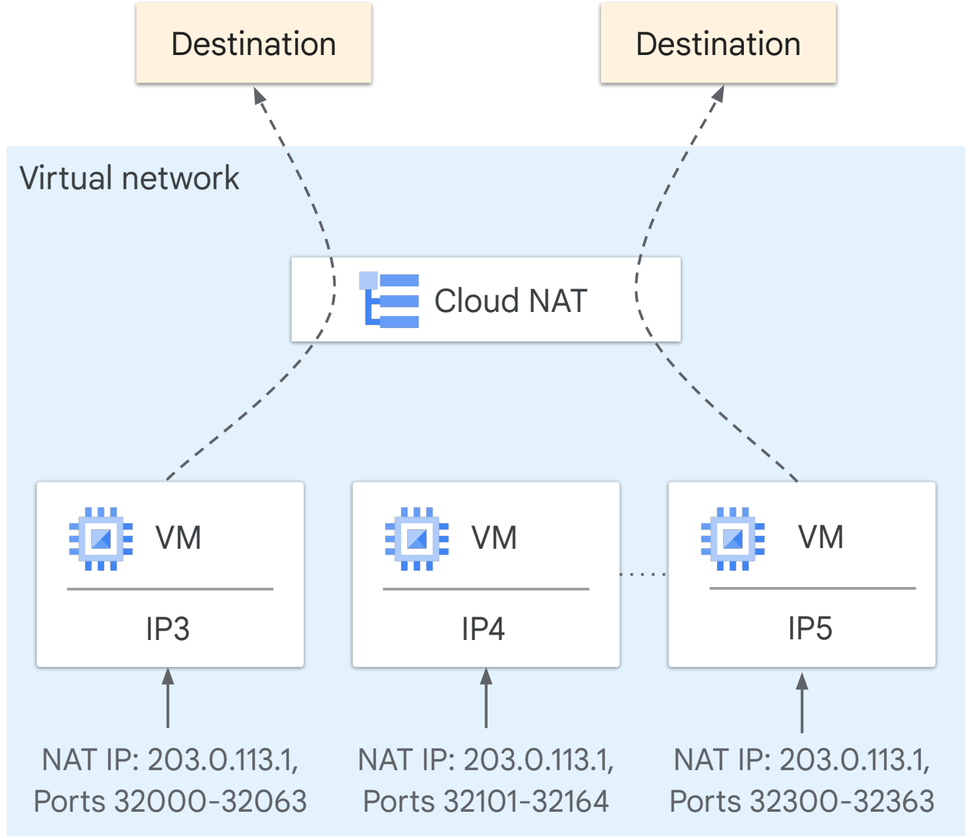


- VPC Flow Logs
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Cloud NAT overview

- It allows Google Cloud Compute workload with no external IP to send packets to the internet.
- It's a fully managed, proxyless NAT service in Andromeda.
- These are some of its benefits:
  - Reduces the need for individual VMs to each have external IP addresses.
  - Automatically scales the number of NAT IP addresses that it uses.
  - Is not dependent on a single physical gateway device.



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Cloud NAT logging

- A NAT log is created when:
  - A network connection using NAT is created.
  - A packet is dropped due to port unavailability
- It lets you log NAT **connections** and/or **errors.**
  - TCP and UDP traffic only.
  - 50-100 entries per second, per vCPU.

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In this section, you explore

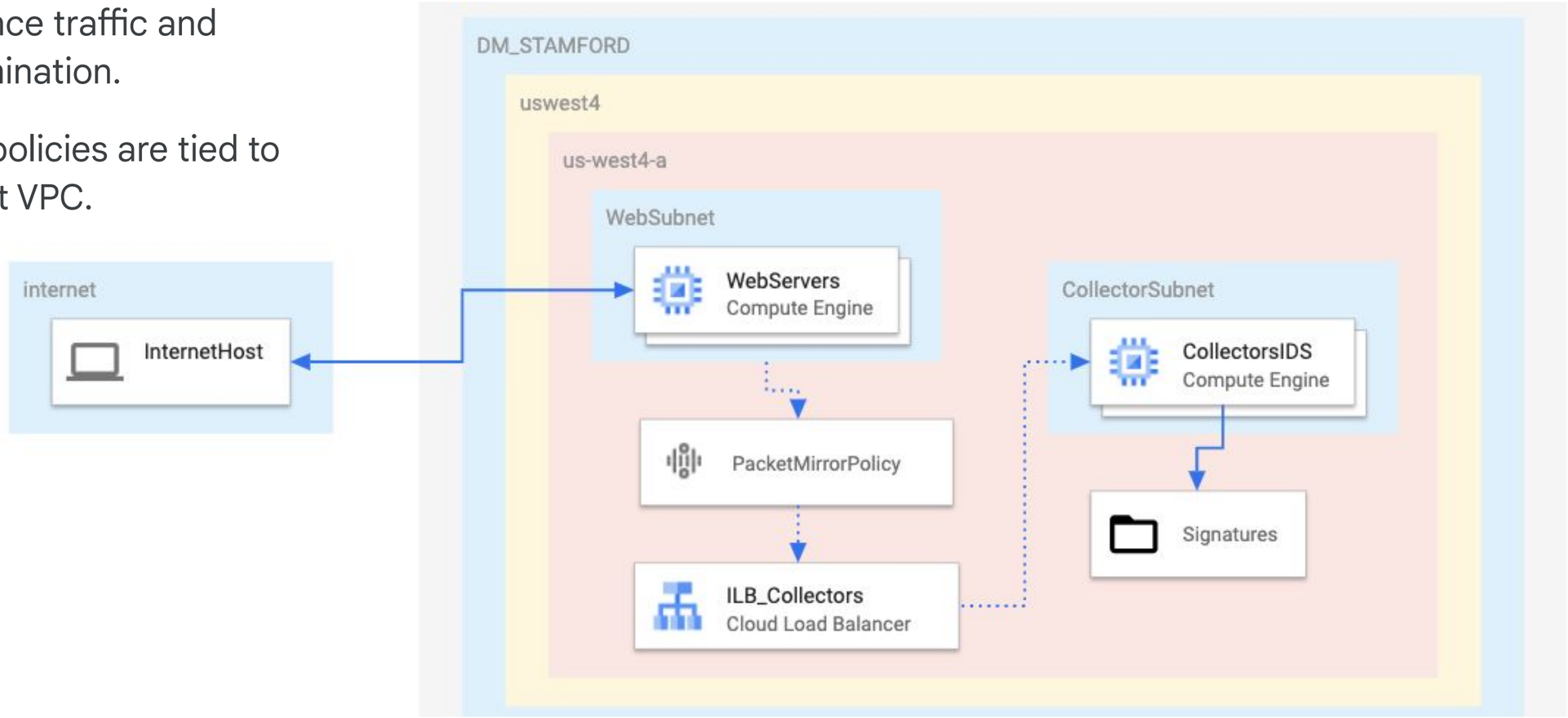


- VPC Flow Logs
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Packet Mirroring: Visualize and protect your network

- Clones VPC instance traffic and forwards for examination.
- Packet Mirroring policies are tied to workloads and not VPC.



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Packet Mirroring: Overcoming bandwidth limitations

- Packet Mirroring consumes the egress bandwidth of the mirrored instances.
  - Use filters to reduce the bandwidth on mirrored instances.
  - Filters can be based on protocol, IP ranges, traffic directions, etc.
  - The current maximum of filters for Packet Mirroring is 30.

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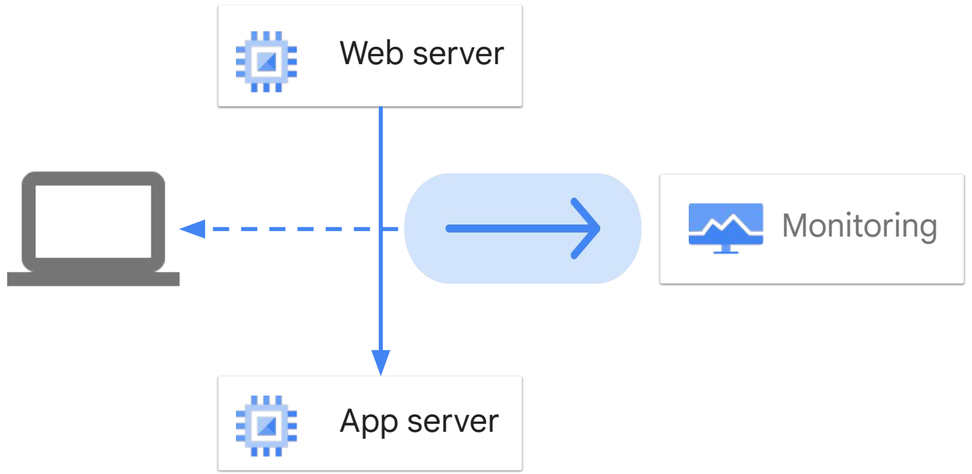
Packet Mirroring: Use cases

- Network and application **monitoring**:
  - Maintain integrity of deployment.
  - Fix packet loss issues.
  - Fix reconnection and latency issues.
- **Security and compliance**:
  - Intrusion detection systems.
  - Deep Packet Inspection engines.
- **Forensics:**
  - Collect, process, and preserve network forensics.

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Monitoring Packet Mirroring

- Use metrics to verify that instances are being mirrored as intended.
  - `Mirrored Packets count`
  - `Mirrored Bytes count`
  - `Dropped Packets count`
- Monitoring can also spot where packet mirroring shouldn't be happening.



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In this section, you explore



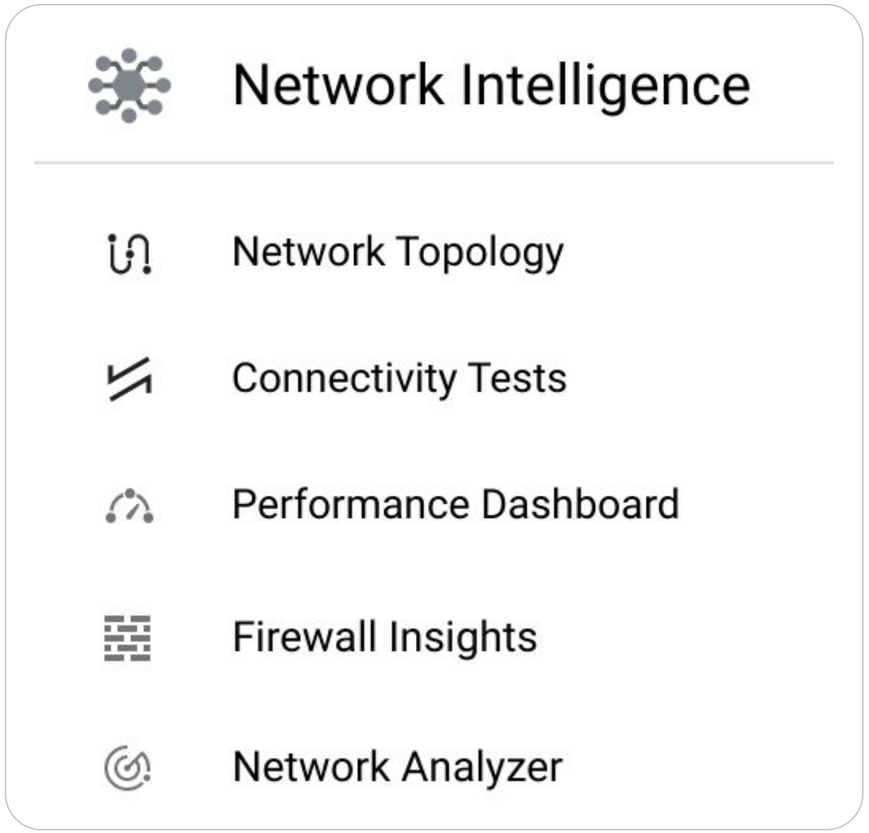
- VPC Flow Logs
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Network Intelligence Center

Centralized network monitoring and visibility

- **Network Topology**: View VPC topology and associated metrics.
- **Connectivity Tests**: Evaluate connectivity to and from VPC resources.
- **Performance Dashboard**: Provides VPC packet loss and latency metrics.
- **Firewall Insights**: Provides visibility into firewall usage and configuration issues.
- **Network Analyzer**: View network and service issues, insights, and recommendations.



Network Topology

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Topology

Entities

- External clients
- Load balancers
- Instances
- NAT gateways
- Network peering

Project aggregation



View flows in BigQuery

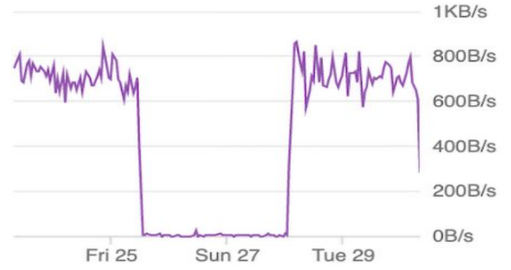
```
select * from [YOUR_TABLE].compute
```

1h 6h 1d 7d 30d

Traffic

EMEA ▶ us-central1

6 sum 1 hr interval (rate)



EMEA ▶ us-central1 (ingress_bytes_count): 606.1B/s

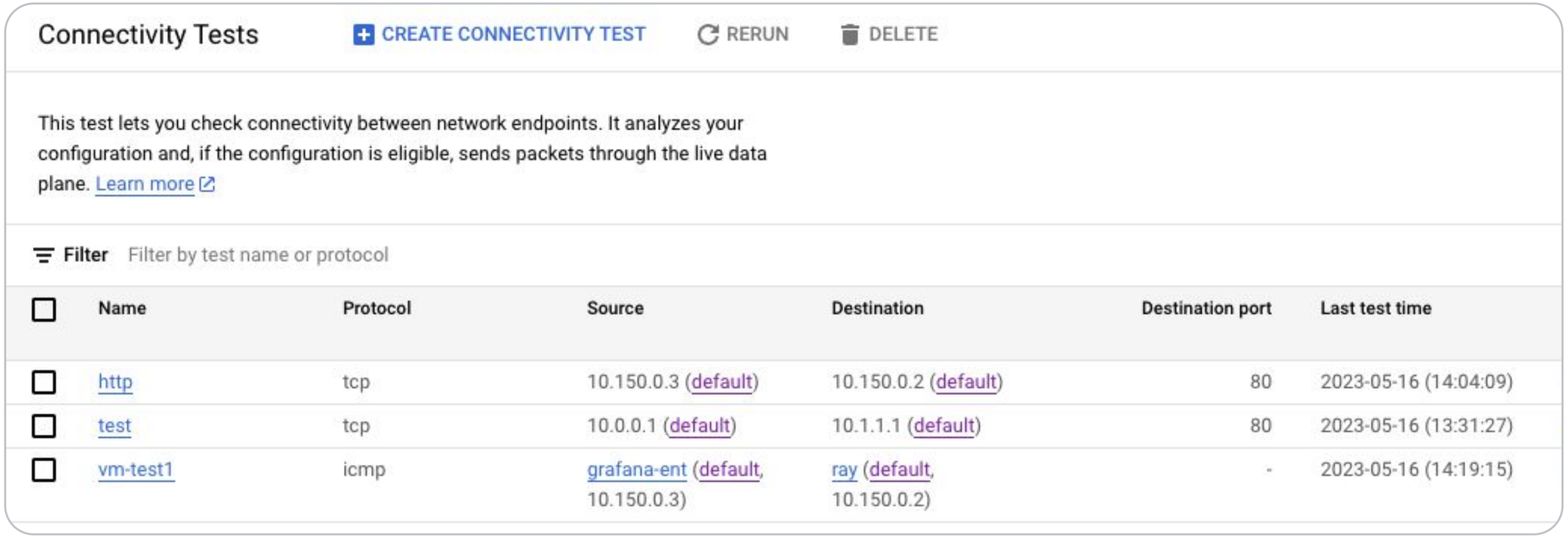
Traffic

us-central1 ▶ EMEA

6 sum 1 hr interval (rate)

Google Cloud

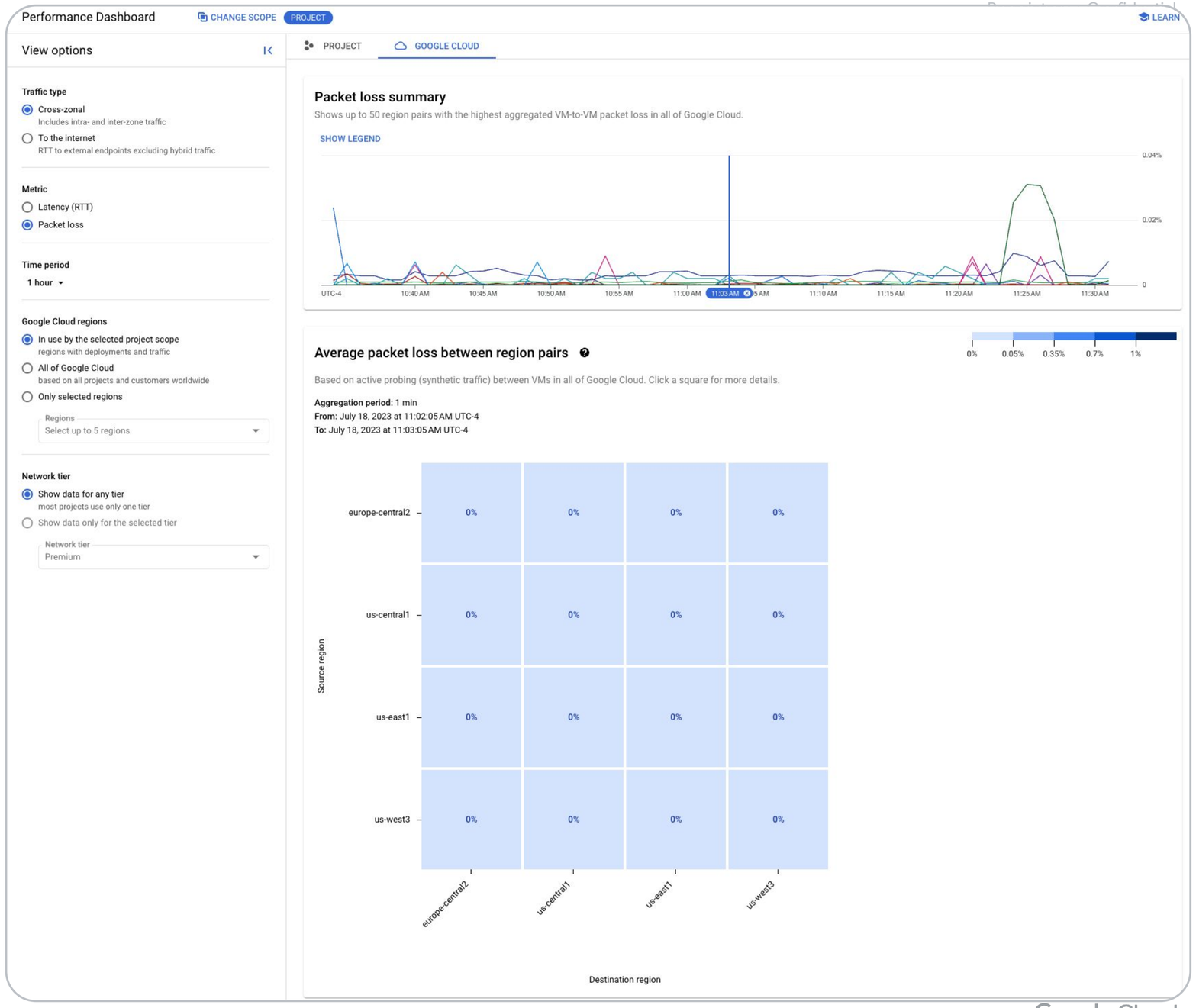
Connectivity Tests



- Quickly diagnose connectivity issues and prevent outages.
- Verify the configuration change effect to help prevent outages.

Performance Dashboard

Packet loss metrics aggregated across zones

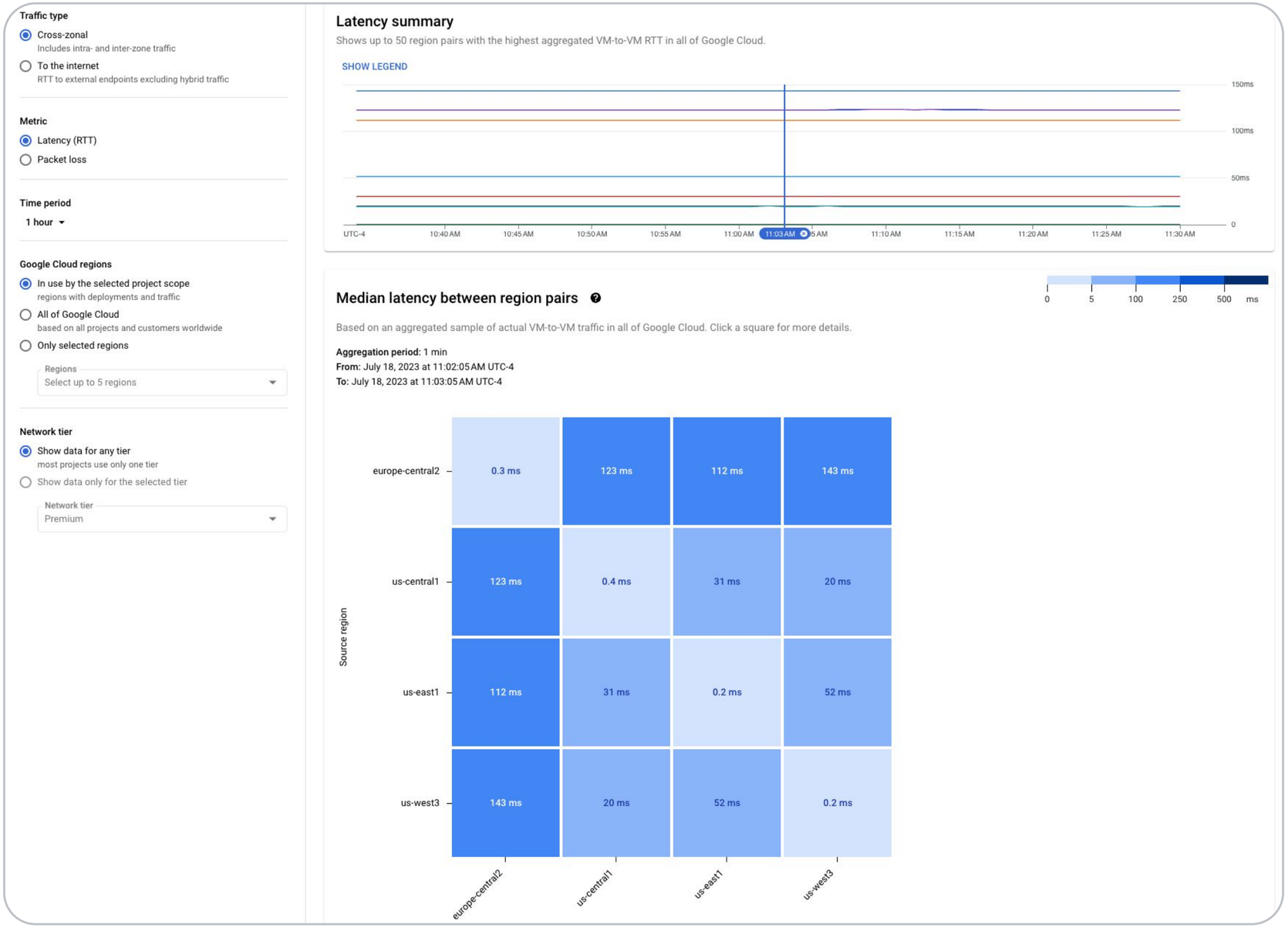


Google Cloud

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Performance Dashboard

Median Latency summaries aggregated across zones

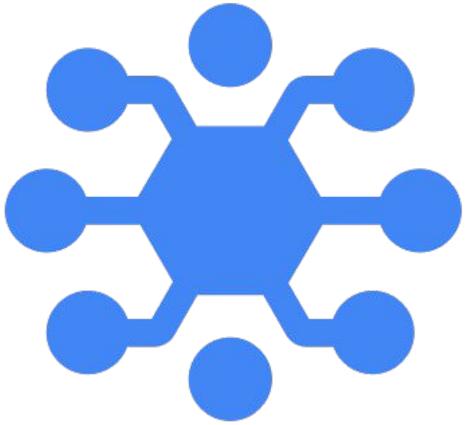


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Firewall Insights helps you better understand and safely optimize your firewall rules



Cloud Monitoring - Metrics



Network Intelligence Center
Firewall Insights



Recommender - Insights



Monitoring

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Firewall Insights let you analyze how your firewall rules are used

- Verify that firewall rules are being used in the intended way.
- Verify that firewall rules allow or block their intended connections.
- Perform live debugging of connections that are inadvertently dropped.
- Discover malicious attempts to access your network.

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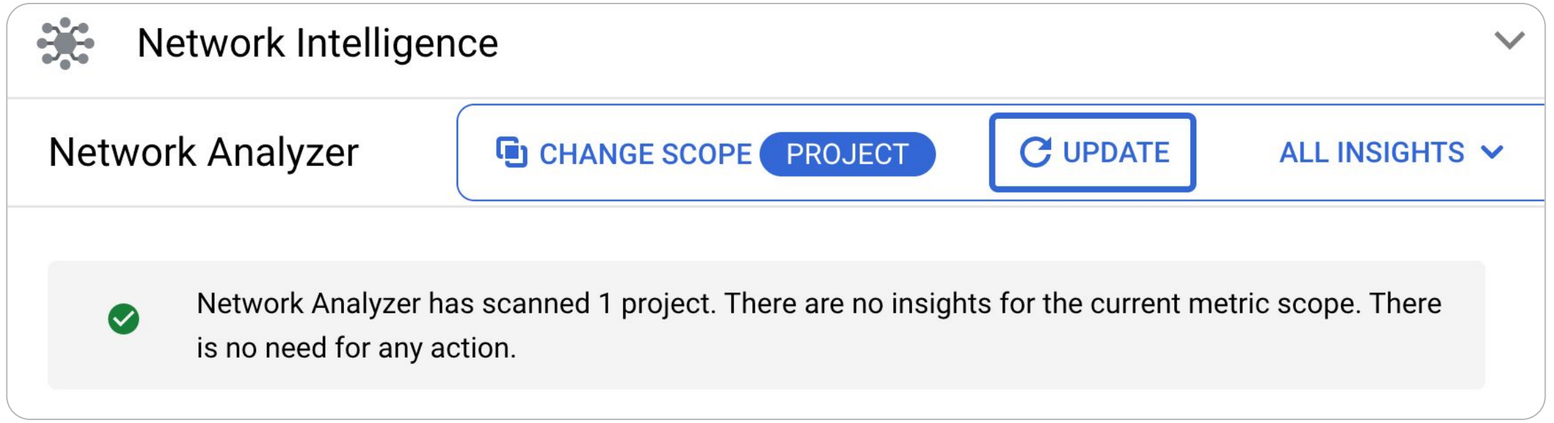
Firewall Insights provides analysis about the firewall rule configuration

- Identify firewall misconfigurations.
- Identify security attacks.
- Optimize firewall rules and tighten security boundaries.

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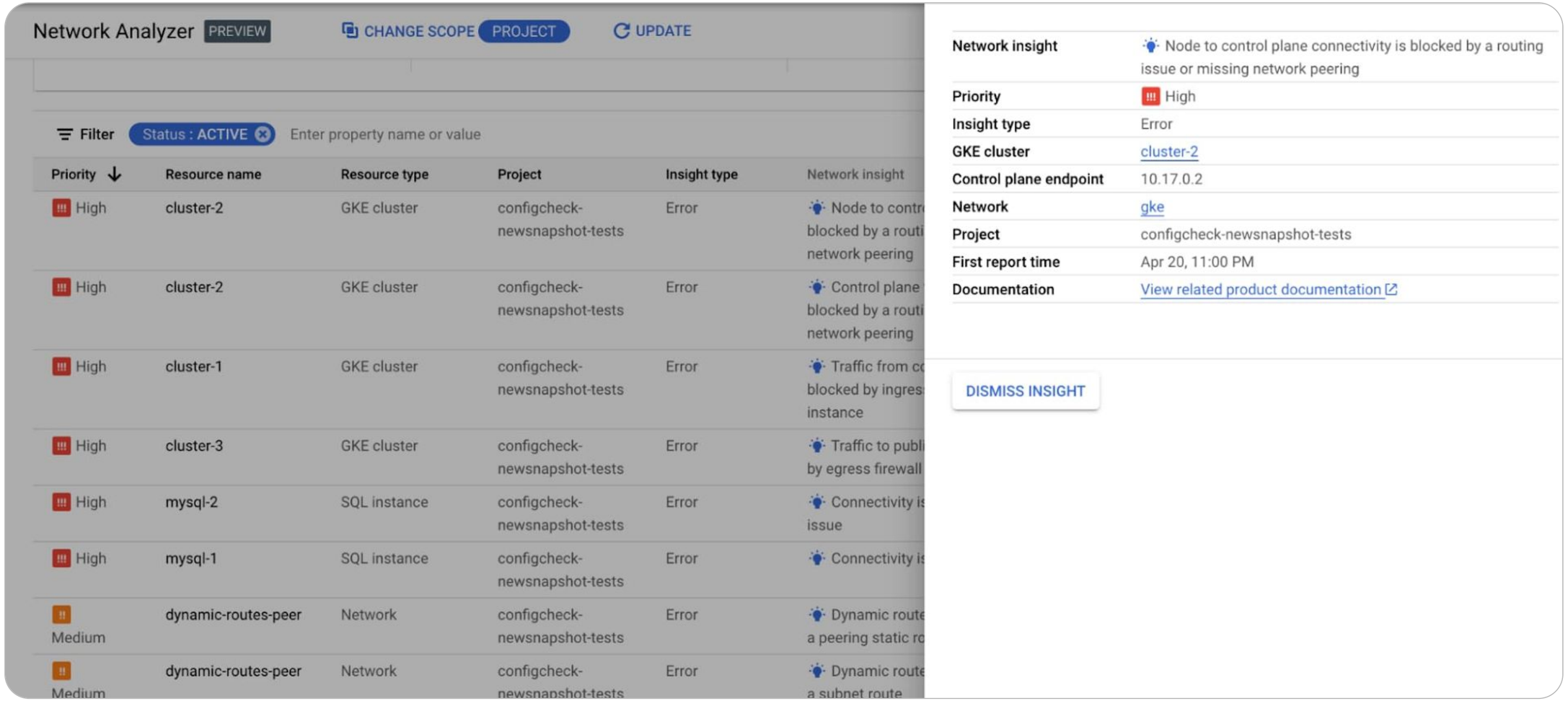
Network Analyzer

- Automatically monitors your VPC network configurations and detects misconfigurations and suboptimal configurations.
- Provides insights on network topology, firewall rules, routes, configuration dependencies, and connectivity to services and applications.
- Identifies network failures, provides root cause information, and suggests possible resolutions.



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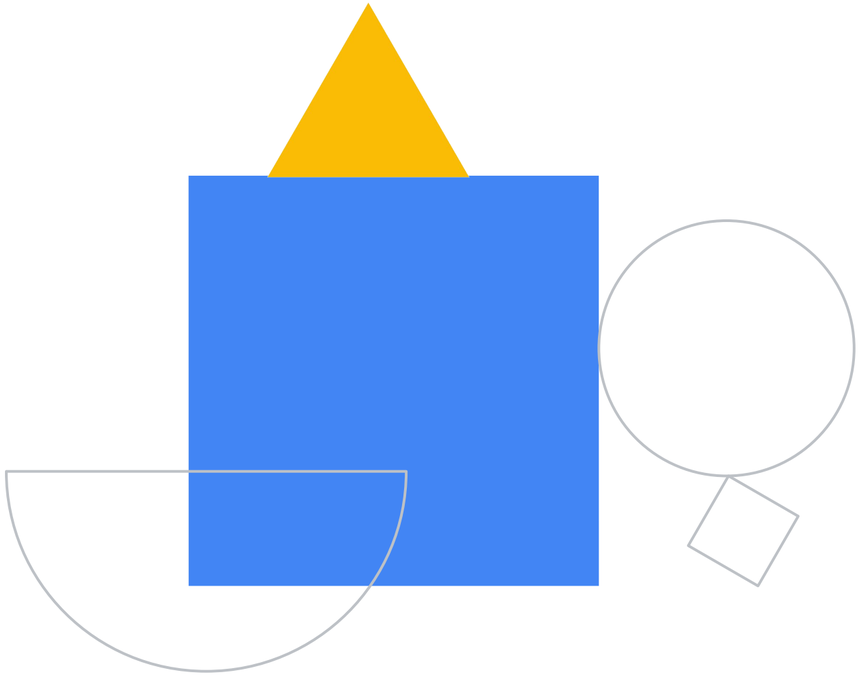
Insights help surface issues and identify the root cause



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Lab Intro

Analyzing Network Traffic with VPC Flow Logs (optional)



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Knowledge Check



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Quiz | Question 01

Question

What logs help you monitor a network by recording a portion of network flows sent and received by VM instances (including GKE nodes)?

A. VPC Flow logs

B. Firewall logs

C. Load balancer logs

D. Cloud NAT logs

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Quiz | Question 01

Answer

What logs help you monitor a network by recording a portion of network flows sent and received by VM instances (including GKE nodes)?

A. VPC Flow logs



B. Firewall logs

C. Load balancer logs

D. Cloud NAT logs

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Quiz | Question 02

Question

What is one of the disadvantages of using packet mirroring?

A. Increased latency

B. Expensive

C. Bandwidth consumption

D. No disaster recovery

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Quiz | Question 02

Answer

What is one of the disadvantages of using packet mirroring?

A. Increased latency

B. Expensive

C. Bandwidth consumption 

D. No disaster recovery

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Quiz | Question 03

Question

What lets you perform a live debugging of connections that are inadvertently dropped?

A. Firewall Rules Logging

B. Firewall Insights

C. Load Balancer logs

D. Cloud NAT logs

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Quiz | Question 03

Answer

What lets you perform a live debugging of connections that are inadvertently dropped?

A. Firewall Rules Logging

B. Firewall Insights 

C. Load Balancer logs

D. Cloud NAT logs

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Recap

01 Collect and analyze VPC Flow Logs, Firewall Rules Logging, load balancer logs, and Cloud NAT logs.

02 Enable and monitor Packet Mirroring.

03 Explain the capabilities of the Network Intelligence Center.



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